



Expansion Analysis with SQL

Solve Business Problems with SQL





Overview

The business aims to expand by opening three coffee shops in India's top three major cities. Since its launch in January 2023, the company has successfully sold its products online and received an overwhelmingly positive response in several cities. As a data analyst, your task is to analyze the sales data and provide insights to recommend the top three cities for this expansion.





Key Business Questions Solved

☐ Easy Level

#	Question	SQL Concept
1	How many people in each city consume coffee? (25% of population)	Aggregation, Arithmetic
2	What is the total revenue from coffee sales in Q4 2023?	EXTRACT, WHERE, SUM





#	Question	SQL Concept
3	How many units of each coffee product have been sold?	LEFT JOIN, COUNT, GROUP BY
4	What is the average sales amount per customer in each city?	JOIN, DISTINCT COUNT, Division
5	List cities with their populations and estimated coffee consumers	CTE, JOIN





☐ Medium Level

#	Question	SQL Concept
6	What are the top 3 selling products in each city?	DENSE_RANK, PARTITION BY, Subquery
7	How many unique customers in each city purchased coffee products?	JOIN, IN filter, COUNT DISTINCT
8	What is the average sale and average rent per customer in each city?	Multiple CTEs, JOIN, ROUND





☐ Hard Level

#	Question	SQL Concept
9	What is the monthly sales growth rate per city?	LAG, CTE, Window Functions, Percentage Calculation
10	Market Potential Analysis – Identify top 3 cities for expansion	All concepts combined





Key Observations

1. **Pune** shows the highest average sale per customer (₹24,197), indicating strong purchasing power.
2. **Delhi** has the largest estimated coffee consumer base (7.7 million), offering the highest customer acquisition potential.
3. **Jaipur** has the lowest average rent per customer (₹156), making it cost-efficient.
4. **Bengaluru** has high consumer potential but significantly higher rent, reducing profitability.
5. **Mumbai** was excluded due to very high average rent per customer.



Top 3 Recommended Cities

PUNE

1. Highest average sale per customer (₹24,197).
2. Highest total revenue based on historical data.
3. Lowest average rent per customer.

Verdict: Best overall city for a new outlet.

DELHI

1. Highest estimated coffee consumers (7.7 million).
2. Highest total active customers (68).
3. Affordable average rent per customer (₹330).

Verdict: Best city for maximum customer reach.

JAIPUR

1. Highest number of current customers (69).
2. Lowest average rent per customer (₹156).
3. Good average sale per customer (₹11,600).

Verdict: Best cost-efficient expansion option.

